

OPINION

A physiological pragmatic view of the philosophical debate on freedom to choose¹Shimon Marom *Technion–Israel Institute of Technology, Haifa, Israel*

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A fleeting curiosity, driven by external events, led me to reflect on the problem of free will. As a hard-nosed physiologist, stepping into the philosophical quagmire of the concept felt far removed from my intellectual comfort zone. Initially, I hoped that exploring the topic would provide clarity and help me articulate a coherent position. I was mistaken. From the outset I found myself increasingly perplexed. What, indeed, was my stance? Was I a dualist? A materialist dualist? A property dualist? A soft materialist? A physicalist? A reductive physicalist? A non-reductive physicalist? An idealist? A functionalist? An emergentist? An extreme emergentist? Or perhaps a compatibilist? A libertarian? A phenomenologist?

There is a story about a Hassidic Rabbi who once pondered a curious question: where should he place his long beard when he went to sleep – above the blanket or beneath it? Typically, this was not an issue, and he would fall asleep with ease. However the night he focused on the dilemma, he struggled. He tried placing the beard above the blanket, then below, and kept switching until, exasperated, he threw off the blanket altogether and fell asleep immediately.

Therefore I have chosen to ‘throw off the blanket’ and engage the topic as a physiologist, liberated from the intricacies of pure philosophical analysis. For decades physiologists and neurocognitive psychologists have sought to illuminate the controversial and topical philosophical problem of free will and its mechanisms, pursuing two primary directions. The first focuses on exploring determinism *versus* non-determinism in brain mechanisms, whereas the second tackles the issue of mental causation, inspired by Libet’s classic study (1983) and its various extensions. Despite advancements in experimental methodologies, the current evidence remains insufficient in addressing either of these arenas (Delnatte et al., 2023; Haggard, 2019).

Here I offer an approach to the subject from a distinctly Jamesian perspective. Specifically, my aim was to explore how physiological facts might inform the discourse on free will, guided by Jamesian pragmatism: ‘The pragmatic method in such cases is to try to interpret each notion by tracing its respective practical consequences’ (James, 1907[1987]). This essay therefore does not seek to argue for or against the existence of free will. Instead, it examines whether and how physiological facts can constrain or shape philosophical discussions on the matter, with a particular emphasis on the role of memory inherent in physiological systems.

To refine the focus I shifted the analysis from free will to free choice. Free choice can be understood as a tangible expression of free will, manifesting through observable decisions. By exploring free choice I aim to provide a concrete lens through which the more abstract concept of free will can be examined.

Ultimately I argue that one’s philosophical stance on free will cannot be grounded in physiological evidence: physiology does not constrain the debate on the freedom to choose, let alone the broader question of freedom to will. The physiological substrate is a structure deeply shaped by its history – the cumulative sum of its life experiences – which may support arguments for determinism. Yet this same substrate is complex and adaptable enough to accommodate the realization of free choice. The reasoning behind this conclusion can be distilled into the following tenets:

Tenet I (underlying assumption)

I assume that a person possesses a certain degree of freedom to select to act one way or another: freedom that reflects, or expresses, a personality structure that is ‘historical’ (hereditary, developmental, environmental). Thus one’s personality does not limit the freedom to decide: the freedom to express one’s personality – this freedom is the essence of the liberty to act one way or another. Action entails movement (or secretion), and thus freedom to act is inherently limited by various constraints of different categories: the physical dimensions and degrees of freedom of the body, relationships with significant subjects and objects, societal and cultural norms, natural laws and more. Together, these factors restrict the freedom to express one’s personality and limit the realization of free choice. Of course this is not all or none, and it is certainly highly circumstance dependent. This aligns with a Hobbesian compatibilist stance, which reconciles free will with determinism by defining freedom as the absence of external constraints, rather than the absence of causation (Hobbes, 1651[1885]). Be that as it may this is a human being – free to choose, to express her personality, to an extent.

Unlike a human being the physiological components of the body – including the brain, brain cells and their constituent proteins – are not individuals; they cannot ‘choose’. Over the years I have regularly measured the electrical activity of networks, cells and membrane proteins. I do not know what it means for a neuron or group of neurons to ‘choose’ or be ‘free’. They are in state *A* or state *B* or any other state, and they certainly do not ‘want’ anything.

Therefore I feel comfortable in the company of those who believe that discussing the concept of free choice, or the concept of will, as well as many other philosophical or psychological concepts, in physiological terms is an error, a categorical error (Marom, 2015). Even if, one day, we achieve the holy grail of measuring every bit of data in a person’s brain, correlating every fragment of behaviour and predicting human behaviour perfectly, it will change nothing. It would be surprising and impressive, but it would not change a person’s freedom to express will by an action. Even if one can perfectly predict

¹ Late in 2024, amidst a time of war, I found myself editing this piece, which stemmed from a lecture delivered at a local meeting earlier in the year. It is a statement on human liberty and the often-overlooked freedom embedded in our very physiology. A gentle reminder that even in such times, there remains a choice; a choice that need not include violence.

what I am going to say in the next few minutes, it is my personality that comes to light. There is no connection between an observer's ability to predict what a person will do and a person's process of choosing what to do.

Tenet II (physiological constraints at the temporal domain)

Although discussing the nature of the concept of free choice might be outside the horizon of physiological relevance, it is still interesting to contemplate constraints that the physiological substrate may contribute to the discussion of freeness to choose. Constraints are helpful; they may guide the discussion, distil it.

Let us focus on temporal constraints at three fundamental levels that are universal to nervous system organization: a single protein, a single cell and a network of cells. Is there something in the dynamics of proteins, cells and networks that constrains human free choice?

To examine this question let us set a straw man in place: assume the existence of an entity that 'freely chooses'. This is purely a thought experiment and need not reflect a dualistic stance. I am not positing an actual entity, nor necessarily endorsing any metaphysical assumptions. Instead, this premise serves as a tool to frame a physiological argument. I don't know what or who (if it exists at all) this entity might be – whether it is matter or spirit is irrelevant for our purposes – as long as it possesses everything necessary to allow even the most rigorous readers to feel comfortable considering it as a 'free-willing entity'.

This is our exploratory assumption: there exists, hypothetically, a free-willing entity. We need this assumption to connect our physiological line of argument with the concept of free choice: this entity, to actualize its will, to choose, must interact with physiology – it must engage with some form of actuator to move the body, activate muscles or vocalize its intentions. In short, this hypothetical entity must have the capacity to alter the state of the system, causing it to shift from one state, *A*, to another, *B* – whether through action potentials, synchronous or asynchronous neuronal activity or another physiological mechanism. Without this physiological connection, the free-willing entity would remain a potential unable to be realized in the physical world.

It is natural to conceptualize the state of the system as progressing along a timeline: past states, the current state and a future state

subject to the decision of the free-willing entity. This decision must manifest through stimulation – an energy investment that prompts a change in activity, directing the system from one state to another. Such a formulation allows us to distil physiological questions that invite measurement and may have relevance to the discussion of the freedom to choose.

Here are the questions: What constraints, if any, does the neurophysiological substrate impose on the free-willing entity in the time domain? Can the free-willing entity implement its will (i.e. choose) at any moment? Whenever does it wish? Or does the nervous system limit the free-willing entity in time? If so, to what extent? How?

As soon as we pose the questions this way, we must acknowledge that not every choice is available (physiologically) for the free-willing entity. There are some trivial constraints. For example after a neuron or group of neurons have fired, there is a period during which the probability of reactivating them is low: a recovery period from the previous activity, a 'refractory' period during which the dynamic system returns to its state of readiness.

Note that, seamlessly, we have shifted from talking about the system's state at a specific *time* to discussing a *duration* within which the probability of reactivating the neuron or network is small. Because we are now dealing with duration, we can proceed to ask: what is the length of the time window during which the free-willing entity has no freedom to implement its will? We all know that when it comes to a single neuron, regardless of what we do, it is impossible to cause it to spike again until at least 2–2.5 ms has passed since its last activation. In a network this may be 100–200 ms. But this, as mentioned, is a trivial constraint; I am confident that the free-willing entity can accept a lower bound on the realization of its desires. That is, it can live with the possibility of freely choosing when to act, provided a known and fixed time window τ has passed. This is acceptable.

But who guarantees us that this duration is fixed? Perhaps the duration within which the free-willing entity cannot realize its will depends on the history of activity? And if so, how far back in time does this dependence go? A tenth of a second? A second? Ten or a hundred or a thousand seconds? Perhaps days? Perhaps the time window for activating a neuron or a network fluctuates cyclically? And if so, what is the characteristic time of

these oscillations? Perhaps, indeed, the probability of successfully activating a neuron or neural network depends on the entire history of neural activity? Does that sound far fetched? Not really. Just as with personality at the psychological level, the history of interactions with an environment filled with dynamic objects and subjects – the entire relational history – dictates the structure of the network and the neuron. This history also determines the nature and strength of the connections, the quantities of relevant proteins and so on. In this sense we can say that the state of a neuron or a network of neurons is the sum of their entire life history. Surely this must, in some way, influence the extent to which neurons can respond to the commands of the free-willing entity.

In short, the degree to which the history of activity influences the neural substrate for realizing the freedom to choose is undoubtedly something that those interested in the question of free will must consider.

Tenet III (memory of all things past)

It is both worthwhile and possible to estimate the extent of this dependence on the history of activity through measurement and experimentation. There are tools to do so. The general outline of such an experiment could look like this. *First*, isolate a basic neural object from its environment. For instance, a single sodium channel protein on the neuron's membrane; these proteins, as you may remember, initiate the creation of a spike through their state changes at the population level. Or, a single neuron with all its extensions, isolated pharmacologically from the network to which it belongs. Or, a recurrent neural network with hundreds or thousands of interconnected neurons, where the degree of connectivity can be controlled to some extent. *Second*, expose the neural object to different histories of activity and ask to what extent its ability to respond to future stimuli is influenced by its past activation. The experiment is challenging because it requires working with a biological system isolated from its environment over time, and one must always ensure to separate experimental instability from the effects that reflect genuine system dynamics. But it is a possible experiment. Many such experiments have been done and are being conducted today.

At every level of investigation – whether examining a neural network, a single neuron or the proteins underlying neural

function – we encounter an intriguing pattern: the probability of a system responding to a stimulus depends, to some extent, on its previous states, no matter how far back in time we look (Marom, 2010). Correlations are present across all time scales (beyond a lower bound set by physical constraints like capacitance and neural conduction speed), with the power spectral density inversely proportional to frequency. That is, the lower the frequency (and thus the longer the correlation), the greater its influence on the system's response probability. These correlations span multiple orders of magnitude in time, continuously and without a characteristic time scale, suggesting that even events in the distant past shape current neural behaviour. This type of temporal dependence is known in professional jargon as 'pink noise', '1/f noise', 'fractal noise' or 'long-tail noise'. A consequence of the system's dependence on past states that extends as far back as can be measured, is that it entails a vast temporal richness, an array of possible activity patterns arising from the system's history.

This result is not unique to the nervous system; it is a common phenomenon in nature. In the human body it is observed in various contexts, such as the distribution of fluctuations in sensory systems and EEG recordings. Surprisingly, it also appears in areas where we might not intuitively expect it, such as modulations in heart rate.

I am not aware of a universal theory explaining the origin of such distributions, although perhaps one exists. But there are certainly explanations for their origin in specific systems. Some particularly elegant results have been achieved in this field.

Tenet IV (interpretation)

There are two different aspects of interpretation, seemingly contradictory, reminiscent of the famous phrase that has been repeatedly mentioned in Jewish-oriented texts about free will: 'Everything is foreseen, yet freedom of choice is granted' (Rabbi Akiva).

On one hand, all is foreseen, as distributions with long-term correlations suggest a certain degree of predictability in the system's future state. The activity of all physiological components – whether

at the level of the brain, network, cell or protein – is history dependent 'all the way down' and 'all the way back', with correlations spanning all measurable time scales. Therefore how can we talk about 'freedom' if the probability of the brain and its components' action – just like the probability of human action – is linked to both the near and distant past, to history?

On the other hand, freedom is granted – because broad distributions provide the system with an extraordinary temporal richness, a multitude of possible time patterns, which the free-willing entity can realize. Moreover there are several demonstrations in the literature of the ability to harness the nervous system to optimally respond to a stimulus, if the temporal pattern of the stimulus itself is broadly distributed. In this sense the brain's state at any given moment is a realm of potential, a collection of possibilities. What emerges from this potential, what becomes realized – whether a pattern or a temporal or spatial scale – is shaped by relations with a dynamic environment, both internal and external, a relational context that poses a daunting methodological challenge (Ch. 5 in Marom, 2015).

What I have tried to express in so many words, Antonio Gramsci formulated beautifully, clearly and sharply in his *Prison Notebooks* (Gramsci, 1929[1971]): 'Structure ceases to be an external force which crushes man, assimilates him to itself and makes him passive, and is transformed into a means of freedom'.

Thus the physiological substrate is a structure that reflects its past, the *précis* of its life story, and at the same time is rich enough to allow for the realization of human freedom, to express one's personality. This returns the ball to the psychological/philosophical court; the issue of understanding free will or free choice is not a physiological one.

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